Day 6 Lab Assignments

Please do these assignments:

1- Create a new page that has a textbox, and alert the code of key pressed on a textbox, and which mouse button clicked on it.

2- On the same page, Make a button (Start clock), that showing alert saying "Clock Started" and displays clock with current time in local format (time only without date) in a div updated every second.
   a. Handle the onclick event of the (start clock) button using JavaScript (Not in the input tag).
   b. Stop the clock when user clicks (alt + w) letter, and show alert saying "Clock stopped".

3- Create a page that contains 3 images, and use JavaScript Code (not in HTML tag) to handle onclick event on all of them, to increase a counter when each image is clicked.
   a. Use addEventListner to add new handler for the onclick event listener, to alert the score (the value of the counter), when any image of them clicked.
      i. What's the difference between handling the event using addEventListner() and other methods?
      ii. Will addEventListner() will overwrite the other handler function attached to that button before?
      iii. How does addEventListner() handles the event bubbling in better way?
   b. After 3 seconds (Use setTimeOut), use removeEventListner to remove the last event handler onclick that alerts the user score, and instead of it add new handler (Using addEventListner), that shows new alert saying "Game Over".

4- Create a login form that redirect to another page when user submit the form, and put in this form textbox (user name) and password field (password) [textboxes must have name attribute] and button (login). User name text has default text "Enter your user name here", when user clicks login he will be redirected to another page (the page given in the form action), use get method and on the other page get the user name (from the browser URL) and write on the page welcome + User Name in bold and big font letters.

5- Make a link that on its onClick event opens new advertising window after 3 seconds, and write long paragraphs on it. Also, make another link that closes this page. (Note that the browser may block the opening of the new window; make sure that you allow this page to open popups in your browser).

6- On registration form you make I Day.2, make a check box and label next to it "I agree to Terms and conditions" (Use label for this text), and make the label as a link that when clicked opens new page (html Page that you should create and put it in the same

folder with the assignment) that contains long text and to be self-scrolling vertically (scrolls vertically automatically every half second).

7- On the same registration page, make the textbox of first name that accepts alphabetic characters only and don't accept numbers to be written on it (check for key code in keydown event, and use preventDefault).

8- In assignment (2) stop the clock when user press (w) instead of (alt+w).
   a. Cancel event Bubbling on the page, so on assignment No.2, if the user presses (w) letter in the textbox (the cursor is in the textbox), it alerts the key code only, but don't stop the clock, and stops only if press (w) letter on the document (when the cursor isn't in the textbox).

Task: Catch The Ball Game

Task Description

Create a simple and fun game where the user tries to catch falling balls using a basket controlled by the mouse!

Requirements

Stage 1: Basics (Level 1)

Your game must include:

1. **Game Area:**
   - Fixed playing area (e.g., 600x400 pixels)
   - Attractive colored background
2. **Basket:**
   - Square/rectangle at the bottom of the screen
   - Moves left and right following mouse movement
   - **Required Events:** `mousemove`
3. **Balls:**
   - Circular balls falling from top to bottom
   - Each ball starts from a random position at the top
   - **Required Events:** Use `setInterval` to move balls
4. **Score Tracking:**
   - When ball touches basket → +1 point
   - Display score on screen
5. **Game Over:**
   - If ball reaches ground without being caught → game ends
   - Display "Game Over" message and final score

Stage 2: Enhancements (Level 2)

Add the following features:

6. **Start/Restart Button:**
   - Button to start the game
   - **Required Events:** `click`
7. **Player Lives:**
   - Player has 3 lives
   - Each missed ball → -1 life
   - Game ends when lives run out
8. **Difficulty Levels:**
   - Every 10 points → ball speed increases
   - Number of balls increases
9. **Sounds (Optional):**
   - Sound when catching a ball
   - Sound when missing a ball

Stage 3: Extra Features (Level 3)

For advanced students:

10. **Keyboard Control:**
    - Right arrow → move basket right
    - Left arrow → move basket left
    - **Required Events:** `keydown` or `keypress`
11. **Different Colored Balls:**
    - Red ball = -1 point (avoid it!)
    - Golden ball = +5 points
    - Blue ball = +1 point (normal)
12. **Power-ups:**
    - Red heart falls = +1 life
    - Clock = slow balls for 5 seconds
13. **Leaderboard:**
    - Save top 5 scores in `localStorage`
    - Display them in a table
14. **Pause:**
    - Pause/Resume button
    - Or press Space bar to pause/resume
    - **Required Events:** `keydown` for space

Required Events to Use

You must use **at least** these Events:

- `mousemove` - Move basket
- `click` - Start/Restart buttons
- `keydown` or `keypress` - Keyboard control (optional for Stage 3)
- `load` - Load game
- Timing Events: `setInterval` - Move balls